



# FXD Engineering Project Binder

## Volume 2 - Milestones 26-31

Audited engineering record based on the approved documentation-recovery structure

|                 |                                                                                           |
|-----------------|-------------------------------------------------------------------------------------------|
| Project         | FXD - Intelligent Industrial Fixture Design                                               |
| Repository      | kool1160/fxd-fixture-design                                                               |
| Document Class  | Engineering development history / professional project binder                             |
| Volume Status   | Closed audited record through Milestone 31                                                |
| Audit Basis     | Merged pull requests, tests, CI, project law, architecture, decision, and project records |
| Runtime Changes | None - documentation only                                                                 |

*AI proposes. Engineering validates.*

*The software should think like an experienced fixture engineer.*

# 1. Executive Summary

Volume 2 records the transformation of FXD from a completed deterministic fixture-engineering foundation into a serious local engineering application with real STEP viewing, a unified PySide6/VTK workbench, interactive engineering workflows, controlled branding, real fixture-construction geometry, and the advisory AI Fixture Engineer with guided validation.

## Governing principles

- Source customer CAD remains immutable.
- Deterministic geometry, mathematics, and engineering rules remain authoritative.
- AI is advisory and may not approve a fixture.
- Every recommendation and validation result must remain explainable and traceable.
- Manufacturing practicality and qualified human approval remain mandatory.

# 2. Master Milestone Recovery Tracker

Exact historical test counts are reported only when the controlling repository evidence states them.

| M  | Milestone                                                  | Status / PR  | Scope Summary                                                                                                                                                                                                                                                                                                                     | Testing / Evidence                                                                                                                                                                                                        | Boundary                                                                                                                                                                      |
|----|------------------------------------------------------------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 26 | Complete the Local Engineering Workbench                   | Complete #48 | Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.                                                                                                                                                  | 5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.                                                                           | Render benchmarks were not visible-GUI FPS claims; engineering review only.                                                                                                   |
| 27 | Unify the FXD Engineering Workbench                        | Complete #49 | Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.                                                                                          | 15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.                                                               | Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.                                                 |
| 28 | Expose the Interactive Fixture Engineering Workflow        | Complete #50 | Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.                                                                                        | 34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.                                                              | Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.       |
| 29 | Implement the Desktop UI and Branding System               | Complete #51 | Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.                                                                                                    | 30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.                                   | Branding does not alter engineering rules, source identity, validation, or approval authority.                                                                                |
| 30 | Generate Real Fixture Geometry and Tack/Location Workflows | Complete #52 | Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.           | 16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.                                                                             | No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release. |
| 31 | AI Fixture Engineer and Guided Validation                  | Complete #53 | Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence. | 17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed. | AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.                                                                 |

### 3. Detailed Milestone Records

## Milestone 26 - Complete the Local Engineering Workbench

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #48 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Repaired OCCT one-based triangle indexing and added persistent VTK actors, shaded defaults, navigation, diagnostics, XCAF color preservation, and honest real-geometry fallback.

### Validation and source evidence

5 focused workbench tests and 154 full-suite tests passed; kernel proof, backlog, compileall, diff checks, and controlled STEP evidence passed.

### Known limitations and protected boundaries

Render benchmarks were not visible-GUI FPS claims; engineering review only.

### Completion record

Milestone 26 is recorded as Complete because its controlling implementation was merged through PR #48. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## Milestone 27 - Unify the FXD Engineering Workbench

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #49 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Replaced the launched Tk shell with a PySide6 workbench and supervised native VTK worker, adding engineering tree, properties/findings, layers, review decisions, renderer controls, diagnostics, benchmarks, selection, and visibility.

### Validation and source evidence

15 focused tests and 164 full-suite tests passed; CI contract, full CI, OCP proof, compileall, backlog, schema, diff checks, and Windows acceptance passed.

### Known limitations and protected boundaries

Windows child-window embedding is not cross-platform acceptance; provisional evidence cannot be labeled real source geometry.

### Completion record

Milestone 27 is recorded as Complete because its controlling implementation was merged through PR #49. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## Milestone 28 - Expose the Interactive Fixture Engineering Workflow

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #50 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Exposed process intent, exact OCP face annotations, private tooling metadata, deterministic background analysis, concept comparison, supported corrections, revision restoration, findings review, persistence, and gated evidence export.

### Validation and source evidence

34 focused workflow/workbench tests and 190 full-suite tests passed; compileall, schema, backlog, CI, secret scan, OCP proof, and Windows acceptance passed.

### Known limitations and protected boundaries

Concept evidence remained visibly provisional; no paid runtime AI, supplier scraping, thermal simulation, robot planning, safety certification, or production approval.

### Completion record

Milestone 28 is recorded as Complete because its controlling implementation was merged through PR #50. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## Milestone 29 - Implement the Desktop UI and Branding System

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #51 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Integrated UI and Branding Kit v1.1 with design tokens, palette, QSS, approved assets, semantic widgets, source identity, workflow rail, renderer health, validation/approval gates, responsive docks, and layout persistence.

### Validation and source evidence

30 focused branding/workbench tests and 201 full-suite tests passed; compileall, schema, backlog, launcher, secret scan, OCP proof, full CI, and branding-manifest verification passed.

### Known limitations and protected boundaries

Branding does not alter engineering rules, source identity, validation, or approval authority.

### Completion record

Milestone 29 is recorded as Complete because its controlling implementation was merged through PR #51. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## Milestone 30 - Generate Real Fixture Geometry and Tack/Location Workflows

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #52 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Added deterministic real OCP fixture-construction evidence for full-weld, tack/location, assembly, inspection, checking, rework, automation, and combined fixture purposes with lifecycle, Cleco, holes, tabs, poka-yoke, BOM, nest, sequence, authority, validation, comparison, authoring, and schema-v4 persistence.

### Validation and source evidence

16 focused tests and 224 full-suite tests passed; compileall, backlog, schema, OCP proof, full CI, and Windows real-geometry evidence passed.

### Known limitations and protected boundaries

No structural adequacy, clamp-force calculation, thermal simulation, robot planning, automated nesting, safety certification, weld-procedure approval, or production release.

### Completion record

Milestone 30 is recorded as Complete because its controlling implementation was merged through PR #52. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## Milestone 31 - AI Fixture Engineer and Guided Validation

|        |          |                |     |
|--------|----------|----------------|-----|
| Status | Complete | Controlling PR | #53 |
|--------|----------|----------------|-----|

### Engineering scope and significance

Added the provider-neutral fxd-fixture-proposal-v1 contract, environment-only adapter, deterministic baseline, output quarantine, proposal workflow, intent confirmation, audited recommendations, evidence highlights, guided validation/correction routing, provenance, staleness, tooling evidence, and schema-v5 persistence.

### Validation and source evidence

17 AI-core, 45 Qt/workbench, 14 persistence/export, 36 orientation/interactive/kernel, and 278 full-suite tests passed; compileall, diff, launcher, CI, hosted OCP acceptance, OCP proof, and Windows walkthrough passed.

### Known limitations and protected boundaries

AI never approves fixtures; deterministic validation and human approval remain authoritative and fail closed.

### Completion record

Milestone 31 is recorded as Complete because its controlling implementation was merged through PR #53. The original pull request, commit history, CI evidence, test records, and qualified engineering review remain authoritative.

## 4. Evidence and Validation Record

Controlling evidence consists of merged pull requests, commit history, automated and hosted validation, real-kernel proof, Windows visual evidence where applicable, architecture decisions, project records, roadmap records, and explicit human-review boundaries. Exact missing historical values are not invented.

## 5. Documentation Correction Note

This audited edition replaces the earlier under-filled print layout. It restores dense milestone tracking, engineering scope, validation evidence, limitations, closeout, and content-driven pagination while preserving repository history as the source of truth.

## 6. Volume Closeout

Volume 2 closes at Milestone 31. FXD now has a professional Windows engineering workbench, real source-geometry review, interactive intent and correction workflows, deterministic real fixture-build geometry, governed persistence, and an advisory AI proposal layer subordinate to validation and human authority.

## 7. Controlled Handoff

Milestone 32 begins the next fixture-family expansion phase, starting with governed multi-station weld-fixture synthesis. It belongs to the next binder volume or addendum.

## 8. Future Documentation Rule

- Planning -> Implementation -> Testing -> Documentation.
- Never mark an open or unmerged milestone complete.
- Update controlled binder batches while evidence is fresh.
- Record PR, validation evidence, limitations, and the next handoff.
- Pass results, not noise; never reconstruct missing facts by guessing.